

EPR and optical studies of Mn^{2+} ions in $\text{Li}_2\text{O}-\text{Na}_2\text{O}-\text{B}_2\text{O}_3$ glasses – An evidence of mixed alkali effect

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Abstract

EPR and optical absorption spectra of 0.5 mol% MnO_2 doped $x\text{Li}_2\text{O}-(30-x)\text{Na}_2\text{O}-69.5\text{B}_2\text{O}_3$ ($5 \leq x \leq 25$) glasses have been studied. The EPR spectra exhibit resonance signals characteristic of Mn^{2+} ions. The resonance signal at $g \cong 2.0$ is due to Mn^{2+} ions in an environment close to octahedral symmetry, whereas the resonances at $g \cong 4.3$ and $g \cong 3.3$ are attributed to the rhombic surroundings of the Mn^{2+} ions. The ionic character (A), the number of spins participating in resonance (N), optical band gap energies (E_{opt}) and Urbach energies (ΔE) show the mixed alkali effect (MAE) with composition. The present study gives an indication that the size of alkalis we choose, is also an important contributing factor in showing the MAE. The variation of N with temperature obeys the Boltzmann law. The optical absorption spectra show a single broad band at $\sim 21\,000\text{ cm}^{-1}$ corresponding to the transition ${}^6\text{A}_{1g}(\text{S}) \rightarrow {}^4\text{T}_{1g}(\text{G})$ which exhibits a blue shift with x . The theoretical values of optical basicity (A_{th}) have also been evaluated.

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1. Introduction

The mixed alkali effect (MAE) is one of the most outstanding and poorly understood phenomenon in glass science [1–14]. It has been observed quite generally in alkali modified network oxide glasses, that the substitution of one alkali by another at a total fixed concentration of the alkalis gives rise to non-linear variation of properties. This is most notable in properties related to alkali ion transport. The other dynamical properties also exhibit a more or less pronounced deviation, viz. internal friction, dielectric loss, viscosity and glass transition etc. The phenomenon is described as mixed alkali effect (MAE) and the field has attracted recent attention [11–14].

Although a number of models and theories have been proposed to account for MAE, none of them give satisfactory explanation for this phenomenon. Approaches made to understand MAE in a particular property often have their own implications on other properties and have therefore been critically examined in the literature. Also, the MAE has a bearing on many technological applications of glasses (toughening of glass, control of dielectric properties, etc.) [3,4,7] it is necessary to examine this phenomenon in greater detail.

As a part of our programme on the MAE [15–17], in the present study, we have carried out electron paramagnetic resonance (EPR) and optical studies in mixed alkali borate ($\text{Li}_2\text{O}-\text{Na}_2\text{O}-\text{B}_2\text{O}_3$) glasses doped with a small quantity of paramagnetic impurity MnO_2 . On the otherhand, the spectroscopic investigations such as EPR and optical absorption techniques are valuable to gain insight into the microscopic origin of the MAE. Our main interest is

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to know whether, the MAE is limited to dynamical properties or the MAE is dependent on the nature of the glass itself. We are also interested to know the effect of alkali ions on the spin-Hamiltonian parameters and also to know the site symmetry around Mn^{2+} ions in these glasses. The effect of temperature (123–300 K) on EPR signals has also been studied.

2. Experimental

2.1. Glass preparation

The glass samples studied in the present work (Table 1) were obtained by the classical melt quenching technique. They were prepared by mixing and grinding together appropriate amounts of Li_2CO_3 , Na_2CO_3 , H_3BO_3 and MnO_2 in an agate mortar before transferring to a porcelain crucible. The mixtures were heated in an electric furnace in air at 1223 K for 20 min. The melt was then quenched to room temperature in air by pouring it onto a polished porcelain plate and pressing it with another porcelain plate. Pink colored glasses have been obtained with good optical quality and high transparency. Finally, the vitreous samples were annealed for 3 h at 423 K to relieve residual internal stress and slowly cooled to room temperature. The polished glasses have been used for optical absorption measurements.

2.2. Measurements

Fig. 1 shows the X-ray diffraction pattern of the samples studied in the present work. X-ray diffraction experiments were carried out with a X-ray diffractometer (Scintag USA), using copper K_α radiation. XRD analysis of samples obtained by quenching revealed the absence of crystalline phase in the glasses. They were consequently the subject of an EPR investigation. EPR spectra were recorded on dry and perfectly powdered glass samples of 100 mg each on an ESR spectrometer (JEOL FE1X) operating in the X-band frequency (9.205 GHz) with a modulation frequency of 100 kHz. The magnetic field was scanned from 0 to 500 mT with a scan speed of 62.5 mT/min. The EPR spectrum of $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ powdered sample was also recorded as a reference to calculate the number of spins.

The EPR spectrum of $20\text{Li}_2\text{O}-10\text{Na}_2\text{O}-69.5\text{B}_2\text{O}_3 + 0.5\text{MnO}_2$ glass sample was recorded at different temperatures (123–300 K) using a variable temperature controller

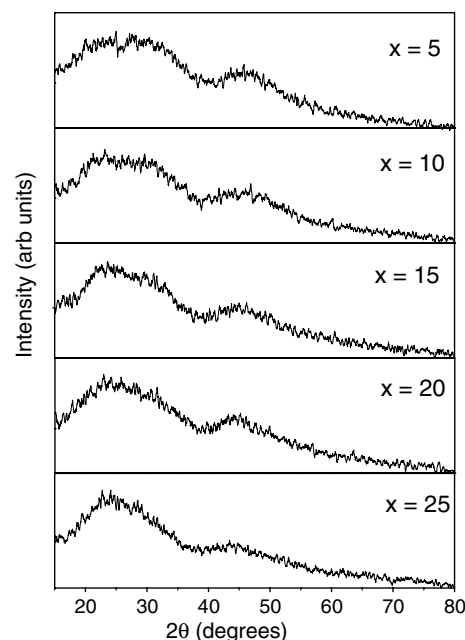


Fig. 1. X-ray diffraction pattern for different mixed alkali borate glasses – $x\text{Li}_2\text{O}-(30-x)\text{Na}_2\text{O}-69.5\text{B}_2\text{O}_3 + 0.5\text{MnO}_2$ ($5 \leq x \leq 25$) at room temperature.

(JES UCT 2AX). A temperature stability of ± 1 K was easily obtained by waiting approximately half an hour at the set temperature before recording a spectrum at each temperature.

The optical absorption spectra of the glass samples were recorded at 300 K on a JASCO (V-570) UV–VIS–NIR spectrophotometer in the range 300–800 nm. The spectra recorded in the ultraviolet wavelength region were used to get the optical absorption edges for the study of optical band gap energies.

3. Results

3.1. EPR studies

The EPR spectra (Figs. 2 and 3) of all the investigated samples (Table 1) exhibit resonance signals due to the Mn^{2+} ($3d^5: {}^6\text{S}_{5/2}$) ions entering the matrix as paramagnetic species. The spectrum of Mn^{2+} ions exhibits a broad resonance at $g \cong 2.0$, shows a six line hyperfine structure (hfs). In addition to this, shoulder around $g \cong 3.3$ and sharp signal at $g \cong 4.3$ were also observed (see Fig. 2). The resonance signal centered at $g \cong 3.3$ is broad, unresolved giving a shoulder like signal. The absorption centered at $g \cong 4.3$ is less intense. Overlapping this signal a narrow, well-resolved line was present. As the starting chemicals used in the present work contain transition metal (TM) impurities (frequently Fe) in trace amounts (10–30 ppm), in undoped glass we also observed the weak signals of Fe^{3+} ions. Nevertheless, as the $g \cong 4.3$ signal is usually observed for Fe^{3+} ions rather than Mn^{2+} ions for trace amounts [18]. We assigned the above signal to accidental

Table 1
Composition of glasses (mol%) studied in the present work

Glass code	Li_2O	Na_2O	B_2O_3	MnO_2
$x = 5$	5	25	69.5	0.5
$x = 10$	10	20	69.5	0.5
$x = 15$	15	15	69.5	0.5
$x = 20$	20	10	69.5	0.5
$x = 25$	25	5	69.5	0.5

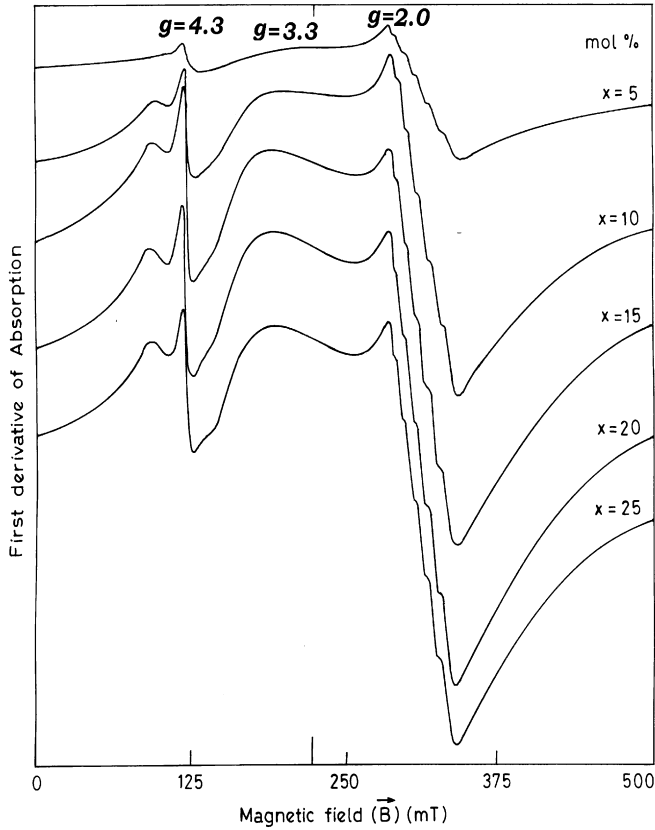


Fig. 2. EPR spectra of 0.5 mol% MnO_2 doped $x\text{Li}_2\text{O}-(30-x)\text{Na}_2\text{O}-69.5\text{B}_2\text{O}_3$ ($5 \leq x \leq 25$) glasses at room temperature.

impurities of Fe^{3+} ($3d^5, ^6\text{S}$) ions. This is confirmed by comparing the linewidth of the signal at $g \cong 4.3$ of Fe^{3+} ions in different glass systems.

3.2. Calculation of spin concentration

The spin concentration (N) can be calculated by comparing the area under the absorption curve with that of a standard ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ in this study) of known concentration. Weil et al. [19] gave the following expression which includes the experimental parameters of both sample and standard.

$$N = \frac{A_x(\text{Scan}_x)^2 G_{\text{std}}(B_m)_{\text{std}}(g_{\text{std}})^2 [S(S+1)]_{\text{std}} (P_{\text{std}})^{1/2}}{A_{\text{std}}(\text{Scan}_{\text{std}})^2 G_x(B_m)_x (g_x)^2 [S(S+1)]_x (P_x)^{1/2}} [\text{Std}], \quad (1)$$

where A is the area under the absorption curve which can be obtained by double integrating the first derivative EPR absorption curve, scan is the magnetic field corresponding to unit length of the chart, G is the gain, B_m is the modulation field width, g is the g factor, S is the spin of the system in its ground state. P is the power of the microwave source. The subscripts ' x ' and 'std' represent the corresponding quantities for Mn^{2+} glass sample and the reference ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$) respectively. The variation of N as a function of alkali content is given in Table 2. It

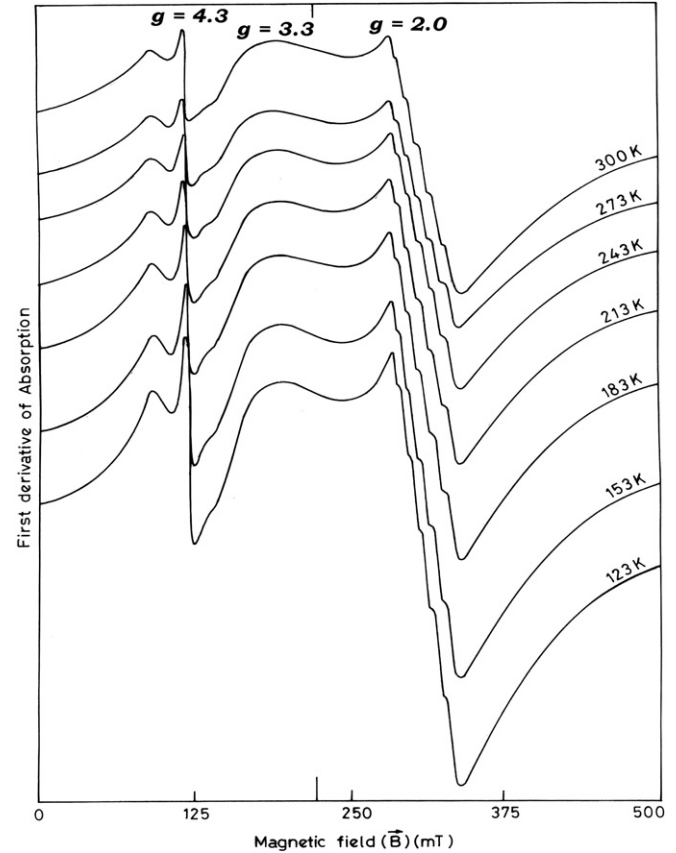


Fig. 3. EPR spectra of $20\text{Li}_2\text{O}-10\text{Na}_2\text{O}-69.5\text{B}_2\text{O}_3 + 0.5\text{MnO}_2$ glass sample as a function of temperature.

is interesting to observe that N increases with x and reaches a maximum around $x = 20$ and thereafter it decreases showing the mixed alkali effect in LiNaB mixed alkali glasses.

The EPR spectrum of $20\text{Li}_2\text{O}-10\text{Na}_2\text{O}-69.5\text{B}_2\text{O}_3 + 0.5\text{MnO}_2$ glass sample was recorded at different temperatures (123–300 K) (Fig. 3) in order to study the temperature dependence of number of spins participating in resonance. Fig. 4 shows a plot of $\log N$ versus reciprocal of absolute temperature ($1/T$) and it is observed that it obeys Boltzmann law.

Table 2

The spin-Hamiltonian parameters, the number of Mn^{2+} ions participating in resonance (N) for 0.5 mol% Mn^{2+} ions in LiNaB glasses at room temperature with x

Glass code (mol%)	g (± 0.001)	Δg	A_{avg} (10^{-4} cm^{-1})	Number of spins (N) per kg ($\times 10^{22}$)
$x = 5$	1.995	−0.0073	83.1	1.04
$x = 10$	1.998	−0.0043	76.3	2.58
$x = 15$	1.998	−0.0043	73.0	3.02
$x = 20$	1.996	−0.0063	76.4	3.44
$x = 25$	1.998	−0.0043	79.6	3.17

The errors in g and A values are ± 0.001 and $2 \times 10^{-4} \text{ cm}^{-1}$, respectively.

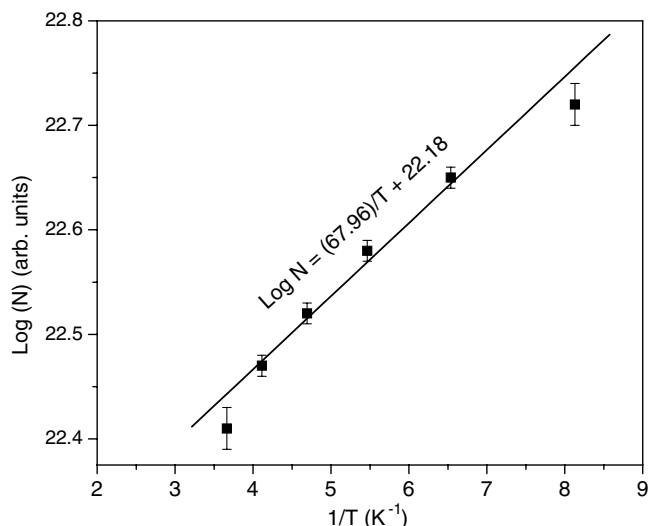


Fig. 4. A plot of $\text{Log } N$ against $1/T$ for $20\text{Li}_2\text{O}-10\text{Na}_2\text{O}-69.5\text{B}_2\text{O}_3 + 0.5\text{MnO}_2$ glass sample. (The solid line is a least square fit with a correlation coefficient of 0.987).

3.3. Optical absorption studies

The optical absorption spectra of $x\text{Li}_2\text{O}-(30-x)\text{Na}_2\text{O}-69.5\text{B}_2\text{O}_3 + 0.5\text{MnO}_2$ ($5 \leq x \leq 25$) glass sample is shown in Fig. 5. From the figure, we can observe that there is only one broad absorption band centered near $\sim 21000 \text{ cm}^{-1}$. This band has been assigned to ${}^6\text{A}_{1g}(\text{S}) \rightarrow {}^4\text{T}_{1g}(\text{G})$ transition of Mn^{2+} ions in a distorted octahedral site.

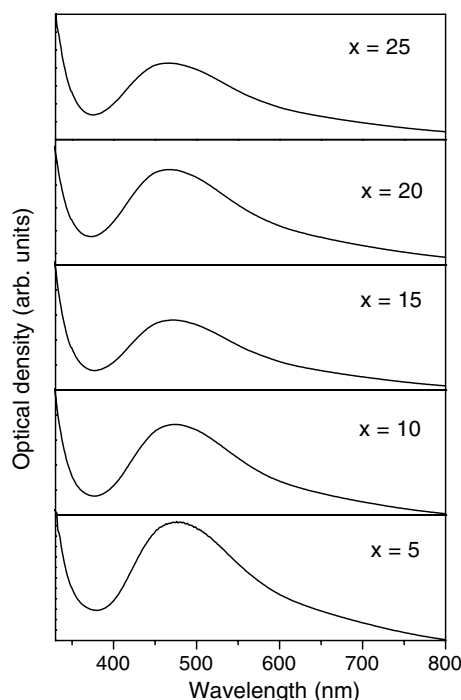


Fig. 5. Optical absorption spectra of different mixed alkali borate glasses – $x\text{Li}_2\text{O}-(30-x)\text{Na}_2\text{O}-69.5\text{B}_2\text{O}_3 + 0.5\text{MnO}_2$ ($5 \leq x \leq 25$) at room temperature.

3.4. Optical band gap energy (E_{opt}) and Urbach energy (ΔE)

The optical band gap and Urbach energies for 0.5 mol% MnO_2 doped $x\text{Li}_2\text{O}-(30-x)\text{Na}_2\text{O}-69.5\text{B}_2\text{O}_3$ ($5 \leq x \leq 25$) glass samples were obtained from their ultraviolet absorption edges. The optical band gap in the amorphous system is closely related to the energy gap between valence and conduction bands [20,21]. In glasses, the conduction band is influenced by the glass forming anions, the cations play an indirect but significant role [20,22]. The Urbach energy gives the width of the tails of localized states within the optical band gap. At the absorption edge, random internal electric fields will dominate the broadening of the exciton levels due to the lack of long range order or presence of defects [20,23]. The least ' ΔE ' i.e. sharp absorption edge suggests that the defects are minimum facilitating long range order.

The main feature of the absorption edge of an amorphous semiconductors is an exponential increase of the absorption coefficient (α) with photon energy ($h\nu$) in accordance with an empirical relation given by Hassan and Hogarth [24]

$$\alpha = \alpha_0 \exp(h\nu/\Delta E), \quad (2)$$

where ' α_0 ' is a constant, ' ΔE ' is the Urbach energy which indicates the width of the band tails of the localized states and ' ν ' is the frequency of radiation. The broadening of the exciton levels at the absorption edge is dominated by the random electric fields due to either the lack of long range order or presence of defects [20,23]. For an absorption by indirect transition, the equation takes the form

$$E_{\text{opt}} = h\nu - (\alpha h\nu/B)^{1/2}, \quad (3)$$

where ' B ' is a constant and ' E_{opt} ' is the band gap. This relation is applied to many oxide glasses [24].

Plots were drawn between $(\alpha h\nu)^{1/2}$ versus $h\nu$ and $\ln(\alpha)$ against $h\nu$ (not shown here). From these plots, the slopes and thereby the optical band gap and Urbach energies have been evaluated in these mixed alkali glasses and are presented in Table 3. The values of optical band gap energies evaluated for $x\text{Li}_2\text{O}-(30-x)\text{Na}_2\text{O}-69.5\text{B}_2\text{O}_3 + 0.5\text{MnO}_2$ ($5 \leq x \leq 25$) mixed alkali glasses at room temperature are presented in Table 3. It is interesting to observe that the optical band gap energies (E_{opt}) and Urbach energies (ΔE) show the mixed alkali effect with composition.

4. Discussion

4.1. EPR studies

Fig. 2 shows the EPR spectra of 0.5 mol% of Mn^{2+} ions in $x\text{Li}_2\text{O}-(30-x)\text{Na}_2\text{O}-69.5\text{B}_2\text{O}_3$ ($5 \leq x \leq 25$) glasses, observed at room temperature. All the glass samples doped with manganese ions show a broad resonance at $g \cong 2.0$ with six line hyperfine pattern, which is a characteristic of Mn^{2+} ions with a nuclear spin $I = 5/2$. Besides the reso-

Table 3

Observed band positions, optical band gap, Urbach energies and optical basicity for 0.5 mol% of Mn^{2+} ions in LiNaB glass matrices with x

Glass code (mol%)	Transition from ${}^6\text{A}_{1g}(\text{S}) \rightarrow {}^4\text{T}_{1g}(\text{G})$ observed peak		Optical band gap energy E_{opt} (eV) ± 0.01	Urbach energy ΔE (eV) ± 0.005	Optical basicity (A_{th})
	Wavelength (nm)	Wave number (cm^{-1})			
$x = 5$	476	21010	3.10	0.333	0.5108
$x = 10$	474	21100	3.28	0.283	0.5073
$x = 15$	472	21185	3.31	0.239	0.5041
$x = 20$	468	21365	3.22	0.265	0.5008
$x = 25$	466	21460	3.17	0.302	0.4976

nance at $g \approx 2.0$, all the glass samples show resonance signals at $g \approx 3.3$ and $g \approx 4.3$ in their EPR spectra. The EPR signal from $g \approx 2.0$ is given by two covered signals: one signal with resolved hyperfine structure (given by the isolated Mn^{2+} ions) and another one signal come from the Mn^{2+} ions involved in dipole–dipole type interactions. The ability to observe the ${}^{55}\text{Mn}$ hyperfine structure has two tangible benefits. (i) it generally allows unambiguous assignments of positions of complex resonance lines to manganese and (ii) the magnitude of hyperfine splitting constant provides a measure of the bonding between Mn^{2+} ions and its surrounding ligands [25,26].

In case of d^5 transition metal ions, it is known that axial distortion of octahedral symmetry gives rise to three Kramers doublets $|\pm 5/2\rangle$, $|\pm 3/2\rangle$ and $|\pm 1/2\rangle$ [27]. Application of Zeeman field lifts the spin degeneracy of the Kramer's doublets. As the crystal field splitting is normally much greater than the Zeeman field, the resonances observed are due to transitions within the Zeeman field split Kramers doublets. The resonances at $g \approx 3.3$ and $g \approx 4.3$ can be attributed to the rhombic surroundings of the Mn^{2+} ions. The resonance at $g \approx 2.0$ is due to the Mn^{2+} ions in an environment close to octahedral symmetry [28] and is known to arise from the transition between the energy levels of the lower doublet, while the resonances at $g \approx 3.3$ and $g \approx 4.3$ arise from the transition between the energy levels of middle Kramer's doublet.

Table 2 gives some typical values of the hyperfine splitting (A) and g values for 0.5 mol% of Mn^{2+} ions in LiNaB glass at room temperature. It is apparent from different average values of A measured from peak to peak, A_{pp} , and trough to trough, A_{tt} , that the individual lines are strain broadened, leading to an asymmetry in the absorption spectrum [29]. The first derivative spectrum as a consequence, shows larger values of A , when measured trough to trough, rather than peak to peak. An overall average was calculated from:

$$A_{\text{avg}} = [(\Delta_{\text{Opp}} + \Delta_{\text{Ott}})/5 + (\Delta_{\text{Mpp}} + \Delta_{\text{Mtt}})/3 + (\Delta_{\text{Ipp}} + \Delta_{\text{Itt}})]/6, \quad (4)$$

where Δ_{Opp} and Δ_{Ott} represent the differences between the first and sixth peak positions, measured peak to peak and trough to trough respectively. Δ_{Mpp} and Δ_{Mtt} represent difference in positions between second and fifth peaks and

Δ_{Ipp} and Δ_{Itt} between second and third peaks. It is interesting to note that, in all of the glass samples, $\Delta_{\text{Opp}}/5 \approx \Delta_{\text{Mpp}}/3 \approx \Delta_{\text{Ipp}}$ and $\Delta_{\text{Ott}}/5 \approx \Delta_{\text{Mtt}}/3 \approx \Delta_{\text{Itt}}$. This indicates that the g -value in each case is well behaved [29].

The magnitude of hfs constant (A) provides a measure of covalency between the Mn^{2+} ions and its ligands. Van Wieringen [25] has noted that the strength of the hyperfine splitting depends on the matrix into which the ion is dissolved and is mainly determined by the electronegativity of the neighbors. This means a qualitative measure of the covalency of the bonding in the matrix which can be determined from the value of A ; the smaller the splitting, the more covalent the bonding of the anion. It was also noted [25] that the g -value for the hyperfine splitting was indicative of the nature of bonding in the glass. If the g -value shows a negative shift with respect to 2.0023, then the bonding is ionic and conversely, if the shift is positive, then the bonding is more covalent in nature. In the present work, from the measured negative shift in the g -value, with respect to 2.0023, it is apparent that the Mn^{2+} ion is in an ionic environment [25]. On this basis, it is observed that the Mn^{2+} ions in LiNaB mixed alkali glasses (with $A \approx 80 \times 10^{-4} \text{ cm}^{-1}$) are quite ionic in nature. The ionic character (A) decreases with x and reaches a minimum around $x = 15$ and thereafter it increases showing the mixed alkali effect.

The number of Mn^{2+} ions participating in resonance at room temperature in $x\text{Li}_2\text{O}-(30-x)\text{Na}_2\text{O}-69.5\text{B}_2\text{O}_3$ ($5 \leq x \leq 25$) mixed alkali glasses have been calculated as a function of x and is presented in Table 2. From Table 2, we can observe that N increases with x and reaches a maximum around $x = 20$ and thereafter it decreases showing the mixed alkali effect in LiNaB mixed alkali glasses. The variation of N may be due to the structural changes with composition and also the modification of boron network with alkali content.

In B_2O_3 glass, the addition of network modifiers (Li_2O and Na_2O) leads to an increase in the coordination number of some of the boron atoms from three to four. It is assumed that the resulting glass is composed of both triangular and tetrahedral units which form a relatively open network with holes between the oxygen atoms of sufficient size to accommodate the Li and Na ions. Thus sufficient non-bridging oxygens are available for good coordination in the broken network. The alkali oxides Li_2O and Na_2O

make available additional weakly bonded O^{2-} for each Mn^{2+} , i.e., Mn^{2+} captures the O^{2-} from Li_2O and Na_2O and this happens at the expense of Li_2O and Na_2O coordination. Na^+ and Li^+ should remain in the neighborhood of the next stronger Mn^{2+} rather than being incorporated separately into the rigid network. The configurations of $Na-O-Mn$ and $Li-O-Mn$ are energetically favored. Replacement of Li_2O by Na_2O causes a smooth change in property–composition curves attaining a value to a maximum or minimum depending on the property.

In our earlier studies [15,17] on sodium–potassium borate ($NaKB$) mixed alkali glass series, we have observed that the number of spins participating in resonance decreases with x reaches a minimum around $x = 20$ and thereafter it increases showing the mixed alkali effect. The exact reverse trend is observed in the present work ($LiNaB$). From these studies, it gives an indication that the ‘size’ of the alkalis play a dominant role. Therefore we propose that the ‘size of the alkalis we choose’ is also an important contributing factor in showing the mixed alkali effect. Our studies further give a clue that, the MAE is not only limited to dynamical properties; it seems it depends on the nature of the mixed alkali glass itself.

Fig. 4 shows a plot of logarithmic dependence of the number of spins ($\log N$) participating in resonance against the reciprocal of absolute temperature ($1/T$). From the plot it is seen that as the temperature is lowered, the spins increases and a linear relationship can be observed between $\log N$ versus $1/T$, obeying the Boltzmann law. The data is least square fit to a straight line $\log N = (67.96/T) + 22.18$ with a correlation coefficient of the fit 0.987. The activation energy thus calculated is found to be 2.159×10^{-21} J (0.013 eV).

4.2. Optical absorption studies

As the Mn^{2+} ions belong to d^5 configuration with 6S ground state, in a cubic crystalline field of low to moderate strength, the five d electrons of Mn^{2+} ions are distributed in the t_{2g} and e_g orbitals, with three in the former and two in the latter. Thus the ground state configuration is $(t_{2g})^3(e_g)^2$. This configuration gives rise to the electronic states, $^6A_{1g}$, $^4A_{1g}$, 4E_g , $^4T_{1g}$, $^4T_{2g}$, and to a number of doublet states of which $^6A_{1g}$ lies lowest according to Hund’s rule. Since, all the excited states of Mn^{2+} ions (belonging to d^5 configuration) will be either quartets or doublets, the luminescence spectra of Mn^{2+} ions will have only spin forbidden transitions.

Fig. 5 shows the optical absorption spectra of $xLi_2O-(30-x)Na_2O-69.5B_2O_3+0.5MnO_2$ ($5 \leq x \leq 25$) glass samples observed in the wavelength region 300–800 nm at room temperature. The absorption spectrum shows single broad band near $\sim 21000\text{ cm}^{-1}$ and is assigned to the $^6A_{1g}(S) \rightarrow ^4T_{1g}(G)$ transition. The exact band positions for individual glasses studied are presented in Table 3. From Table 3, it is interesting to observe that, the band maximum corresponding to $^6A_{1g}(S) \rightarrow ^4T_{1g}(G)$ transition

of Mn^{2+} ions in $LiNaB$ glasses exhibit a slight blue shift with x .

4.3. Optical basicity (A_{th})

The optical basicity of an oxide glass can be measured in terms of the ability to donate negative charge to the ion [30]. Duffy and Ingram [31] proposed that the ideal values of optical basicity can be predicted from the composition of the glass and basicity moderating parameters of various cations present.

The theoretical value of the optical basicity, A_{th} of the glass can be estimated by using the formula [31]

$$A_{th} = \sum_{i=1}^n \frac{Z_i r_i}{2\gamma_i}, \quad (5)$$

where ‘ n ’ is the total number of cations present, ‘ Z_i ’ is oxidation number of the i th cation, ‘ r_i ’ is the ratio of number of i th cation to the number of oxides and ‘ γ_i ’ is the basicity moderating parameter of the i th cation. The basicity moderating parameter ‘ γ_i ’ can be calculated from the following equation

$$\gamma_i = 1.36(x_i - 0.26), \quad (6)$$

where ‘ x_i ’ is the Pauli electronegativity [32] of the cation. The theoretical values of optical basicity (A_{th}) were calculated for all the glass samples and are listed in Table 3. It is found that the optical basicity of the glasses decreases with increase of x .

4.4. Optical band gap energy (E_{opt}) and Urbach energy (ΔE)

Fig. 6 shows the variation of optical band gap energies in $LiNaB$ glasses with x . It is interesting to observe that the band gap energy increases with x up to $x = 15$ reaching a maximum at around $x = 15$ and thereafter decreases showing the mixed alkali effect. The values obtained in the present work are of the same order reported in the literature for borate glasses. Plots were also drawn with $\ln(\alpha)$ against $h\nu$ (not shown in the figure) for the mixed alkali borate glasses in order to calculate the values of Urbach energy. The values of Urbach energy, ΔE , were calculated by determining the slopes of the linear regions of the curves and taking their reciprocals. The Urbach energies obtained for the glasses studied in the present work are also listed in Table 3. Fig. 7 show the variation of Urbach energy with x and it is observed that ΔE , decreases with x and reaches a minimum around $x = 15$ and thereafter increases showing the mixed alkali effect. The small value of Urbach energy in the present study for $x = 15$ suggests that the defects are minimum in this glass sample at $x = 15$.

5. Conclusions

We have shown for the first time that electron paramagnetic resonance and optical absorption techniques are use-

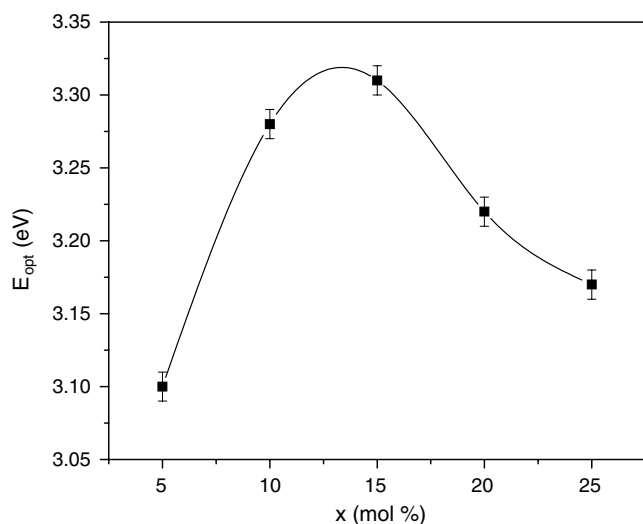


Fig. 6. The variation of optical band gap energy (E_{opt}) for 0.5 mol% MnO_2 doped $x\text{Li}_2\text{O}-(30-x)\text{Na}_2\text{O}-69.5\text{B}_2\text{O}_3$ ($5 \leq x \leq 25$) glasses. (The line is drawn as a guide to the eye).

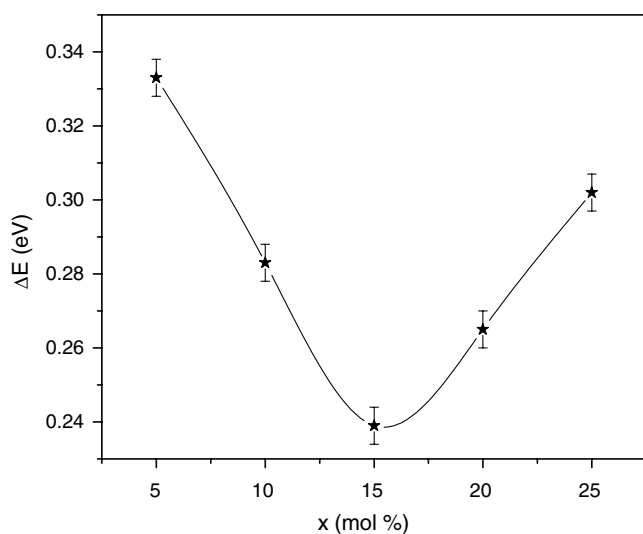


Fig. 7. Composition dependence of Urbach energy (ΔE) for 0.5 mol% MnO_2 doped $x\text{Li}_2\text{O}-(30-x)\text{Na}_2\text{O}-69.5\text{B}_2\text{O}_3$ ($5 \leq x \leq 25$) glasses. (The line is drawn as a guide to the eye).

ful tools to study the mixed alkali effect. The EPR spectra of all the investigated samples exhibit resonance signals characteristic of Mn^{2+} ions. The ionic character (A) decreases with x and reaches a minimum around $x = 15$ and thereafter it increases showing the mixed alkali effect. It is interesting to note that the number of Mn^{2+} ions participating in resonance (N) exhibits the mixed alkali effect. The present study also gives an indication that the size of alkalis we choose, is also an important contributing factor in showing the MAE. Our studies further give a clue that, the MAE is not only limited to dynamical properties; it seems it depends on the nature of the mixed alkali glass itself. The variation of N with temperature obeys the Boltzmann law. The optical absorption spectrum exhibits a sin-

gle broad band and shows a slight blue shift with x . It is interesting to note that the optical band gap energies and the Urbach energies for these glasses show the mixed alkali effect.

The existing theories of MAE have been proposed mainly on the basis of transport properties in mixed alkali glasses, particularly electrical. On the other hand, spectroscopic investigations on mixed alkali effects are very meagre and appropriate theoretical models are not available. The present investigation is a part of our programme to study MAE using spectroscopic techniques and we have shown that the EPR and optical parameters do show mixed alkali effect. Further work is needed with other spectroscopic techniques like Raman, IR etc., so that appropriate theoretical models could be developed. We hope that such studies will draw the attention of theorists.

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References

- [1] R.M. Hakim, D.R. Uhlman, *Phys. Chem. Glasses* 8 (1967) 174.
- [2] J.O. Isard, *J. Non-Cryst. Solids* 1 (1969) 235.
- [3] D.E. Day, *J. Non-Cryst. Solids* 21 (1976) 343.
- [4] M.D. Ingram, *Phys. Chem. Glasses* 28 (1987) 215.
- [5] P. Maass, A. Bunde, M.D. Ingram, *Phys. Rev. Lett.* 68 (1992) 3064.
- [6] G.N. Greaves, K.L. Ngai, *Phys. Rev. B* 52 (1995) 6358.
- [7] K.J. Rao, *Structural Chemistry of Glasses*, Elsevier, Amsterdam–Boston, 2002.
- [8] J. Swenson, A. Matic, C. Karlsson, L. Borjesson, C. Meneghini, W.S. Howells, *Phys. Rev. B* 63 (2001) 132202.
- [9] J. Swenson, S. Adams, *Phys. Rev. Lett.* 90 (2003) 155507.
- [10] A. Bunde, M.D. Ingram, S. Russ, *Phys. Chem. Chem. Phys.* 6 (2004) 3663.
- [11] A. Faivre, D. Viviani, J. Phalippou, *Solid State Ionics* 176 (2005) 325.
- [12] Peibst Robby, Schott Stephan, Maass Philipp, *Phys. Rev. Lett.* 95 (2005) 115901.
- [13] Lammert Heiko, Heuer Andreas, *Phys. Rev. B* 72 (2005) 214202/1.
- [14] Gao Yong, *Chem. Phys. Lett.* 417 (2006) 430.
- [15] R.P. Sreekanth Chakradhar, K.P. Ramesh, J.L. Rao, J. Ramakrishna, *J. Phys.: Condens. Mat.* 15 (2003) 1469.
- [16] Y.C. Ratnakaram, R.P. Sreekanth Chakradhar, K.P. Ramesh, J.L. Rao, J. Ramakrishna, *J. Phys.: Condens. Mat.* 15 (2003) 6715.
- [17] R.P. Sreekanth Chakradhar, K.P. Ramesh, J.L. Rao, J. Ramakrishna, *J. Phys. Chem. Solids* 64 (2003) 641.
- [18] D. Loveridge, S. Parke, *Phys. Chem. Glasses* 12 (1) (1971) 19.
- [19] J.A. Weil, J.R. Bolton, J.E. Wertz, *Electron Paramagnetic Resonance-Elementary Theory and Practical Applications*, Wiley, New York, 1994, p. 498.
- [20] G. Fuxi, *Optical and Spectroscopic Properties of Glasses*, Springer, Berlin, 1992, p. 62.
- [21] C. Dayanand, R.V.G.K. Sarma, G. Bhikshamaiah, M. Salagram, *J. Non-Cryst. Solids* 167 (1994) 122.
- [22] P. Nachimuthu, P. Harikishan, R. Jagannathan, *Phys. Chem. Glasses* 38 (1996) 59.
- [23] E. Davis, N. Mott, *Philos. Mag.* 22 (1970) 903.
- [24] M.A. Hassan, C.A. Hogarth, *J. Mater. Sci.* 23 (1988) 2500.

- [25] J.S. Van Wieringen, Discuss. Faraday Soc. 19 (1955) 118.
- [26] F.D. Tsay, L. Helmholz, J. Chem. Phys. 50 (1969) 2642.
- [27] A. Abragam, B. Bleaney, Electron Paramagnetic Resonance of Transition Ions, Clarendon, Oxford, 1970.
- [28] I. Ardelean, M. Peteanu, V. Simon, S. Simon, M. Flora, Phys. Chem. Glasses 41 (2000) 153.
- [29] J.R. Pilbrow, Bull. Magn. Reson. 9 (1987) 32.
- [30] E. Guedes de Sousa, S.K. Mendiratta, J.M. Machado da Silva, Portugal Phys. 17 (1986) 203.
- [31] J.A. Duffy, M.D. Ingram, J. Inorg. Nucl. Chem. 37 (1975) 1203.
- [32] L. Pauling, The Nature of Chemical Bond, 3rd Ed., Cornell University, NewYork, 1960, p. 93.